Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

Name: Shravan P M
Roll No.: 24110026
Registration No.: 24011101101

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

`https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist`

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$\boxed{x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}}$$

where x is the input image and $\hat{x}$ is its reconstruction.

Execution setup actually used. 10,000 MNIST training images were split into 9,000 for training and 1,000 for validation; the first 2,000 MNIST test images were used for testing. Pixels were scaled to $[0, 1]$ (verified range 0.0–1.0). Seed = 42; TensorFlow 2.20.0 / Keras 3.13.2; Adam (10^{-3}), batch size 128, 20 epochs (30 epochs for the VAE).

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32\times28\times28\times1}.$$

Pixel values should be converted from

$$[0, 255] \to [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28\times28\times1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28\times28\times1}.$$

Students must display:

```
Original image
Reconstructed image
Reconstruction error
```

For the denoising model, display:

```
Original clean image
Noisy image
Denoised reconstruction
```

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.
The encoder maps the input to a lower-dimensional representation:

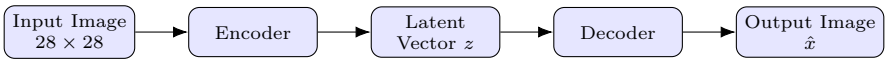
$$z = f_\theta(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_\phi(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2.$$



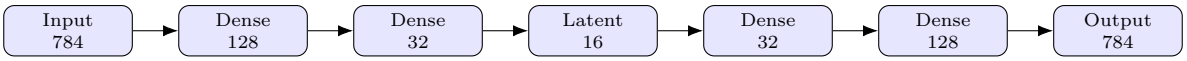
The latent vector is a compressed representation of the input.

7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.
Use the following architecture:

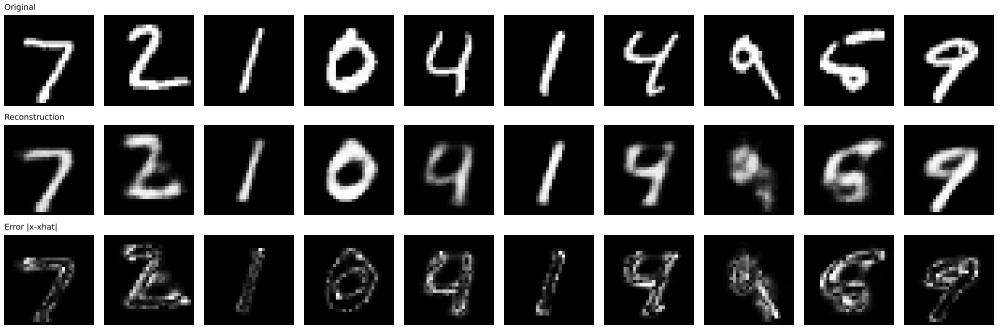


Use ReLU activation in hidden layers and sigmoid activation in the output layer.
Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images
Display at least 10 test images in a grid.
For every image, show:

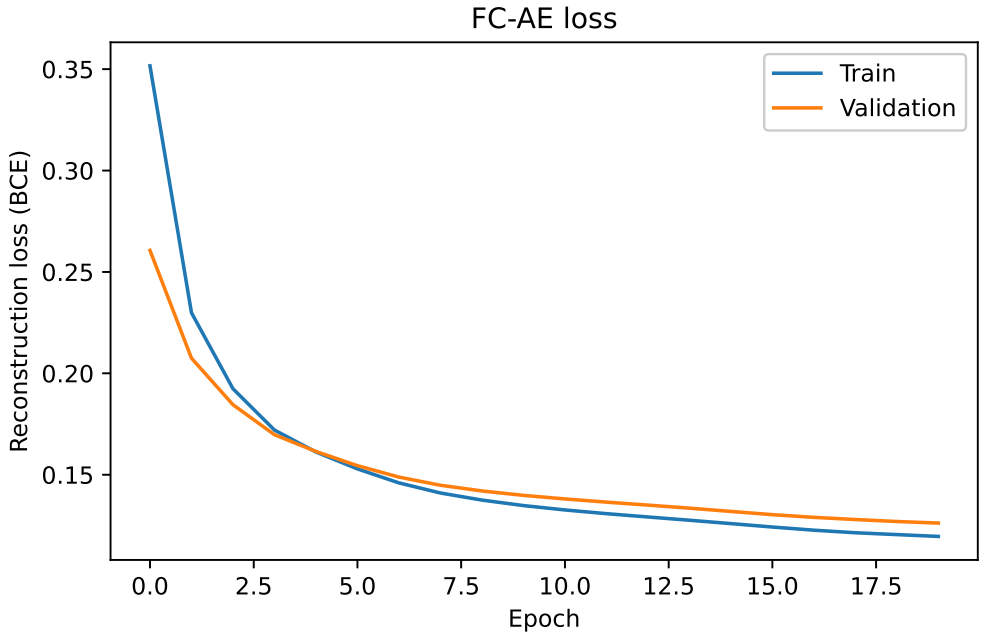


Students must comment on:

The fully connected autoencoder preserves the overall shape and identity of most MNIST digits, particularly simple digits such as 1 and 7. However, fine stroke-level details are lost during reconstruction, resulting in smoother and slightly blurred digit boundaries. More complex digits such as 2, 6, and 4 exhibit greater reconstruction errors, especially around curved regions and thin strokes. Thus, easy samples with simple structures are reconstructed more accurately, whereas difficult samples with complex shapes and variations in handwriting suffer greater information loss.

- **Preservation of digit shape:** The overall structure and identity of most digits are preserved. Simple digits such as 1 and 7 are reconstructed particularly well.
- **Loss of fine details:** Thin strokes, small variations in handwriting, and fine edge details are not reproduced exactly, causing the reconstructed digits to appear simplified.
- **Blurred regions:** Blurring is mainly visible around digit boundaries and curved strokes. Digits such as 2, 6, and 4 show smoother and less-defined edges compared with the original images.
- **Easy vs. difficult samples:** Simple digits with clear structures are reconstructed more accurately, while complex or highly curved digits produce larger reconstruction errors. This indicates that the compressed latent representation makes it more difficult for the fully connected autoencoder to retain complex spatial details.

Plot 2: Training and Validation Reconstruction Loss



- X-axis: Epoch
- Y-axis: Reconstruction loss
- Curves: Training loss and validation loss

Required inference: Determine whether the model converges and whether there is evidence of overfitting.

The training and validation losses decrease and stabilize, indicating that the model converges successfully. The validation loss closely follows the training loss without increasing, showing no significant evidence of overfitting.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|x_i - \hat{x}_i|.$$

Structural Similarity

Calculate SSIM between the original and reconstructed images.

The implementation may use:

`https://scikit-image.org/docs/stable/api/skimage.metrics.html`

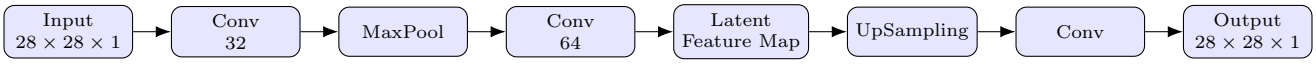
Report:

Metric	Value
Test MSE	0.00256
Test MAE	0.01500
Mean SSIM	0.9739

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

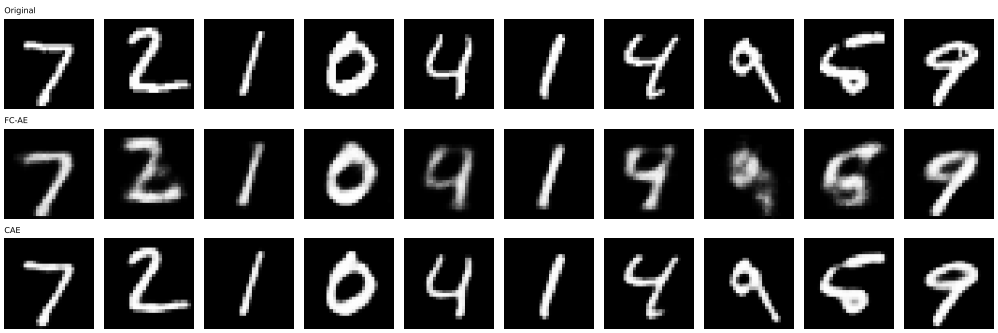
Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02050	0.05533	0.7581	211,040
Convolutional AE	0.00256	0.01500	0.9739	74,497

Plot 3: Reconstruction Comparison

Display:

Original → FC-AE Reconstruction → CAE Reconstruction.



Required inference:

Discuss how preserving spatial locality through convolution affects image reconstruction.

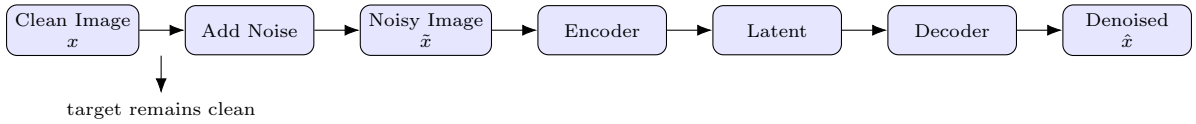
Convolution preserves spatial locality by capturing nearby pixel relationships and local features such as edges and strokes. This allows the CAE to reconstruct digit shapes more accurately, resulting in lower MSE and higher SSIM than the fully connected autoencoder.

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.
The training mapping is:

$$\boxed{\tilde{x} \to \text{Encoder} \to z \to \text{Decoder} \to \hat{x}}$$

where $\tilde{x}$ is a noisy version of the clean image x.



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \qquad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.
Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

Denoising training protocol used. Two denoising CAEs with the same architecture as the CAE were trained: one on Gaussian noise ($\sigma = 0.2$) and one on salt-and-pepper noise ($p = 0.10$); fresh noise was drawn every epoch and the target was always the clean image. Each model was then evaluated on all three noise levels of both noise types (against the clean test image).

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised
For at least 10 images, display:

$$\text{Clean} \quad | \quad \text{Noisy} \quad | \quad \text{Denoised}.$$

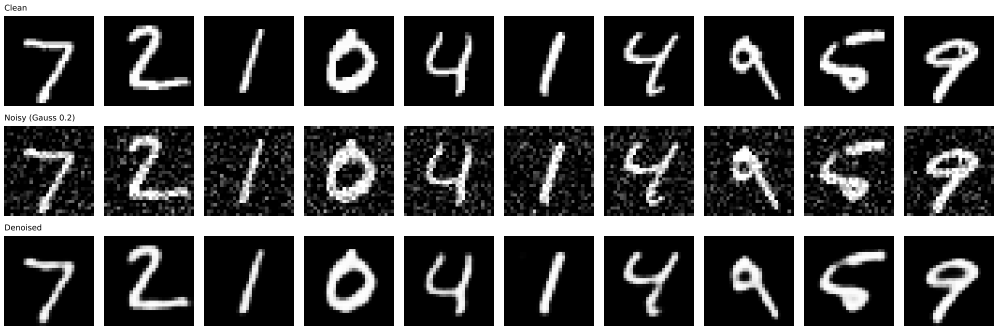
Plot 5: Noise Level vs. Reconstruction Quality
For each noise level, calculate:

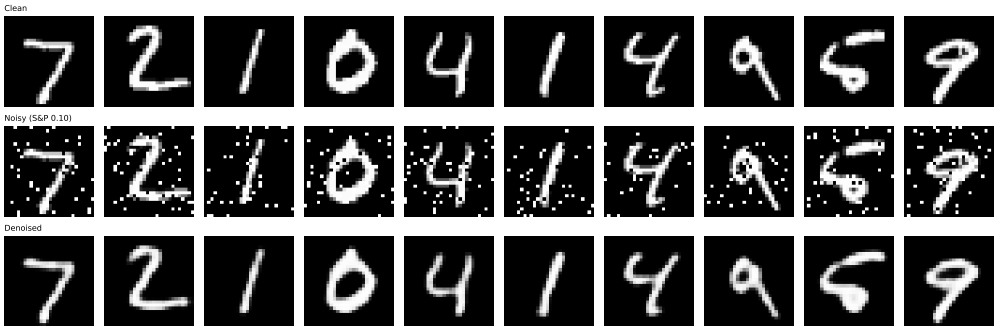
$$MSE, \quad MAE, \quad SSIM.$$

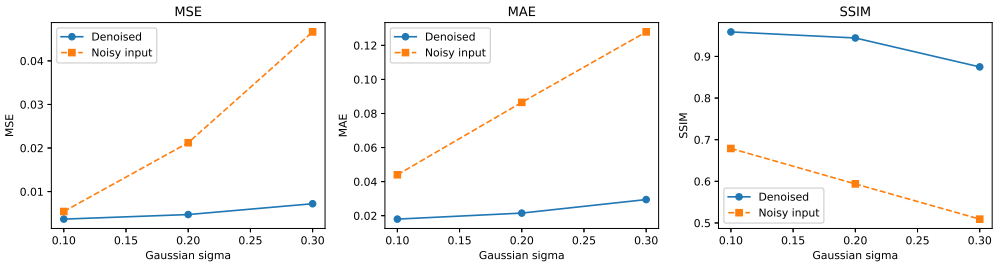
Report:

Noise Level	MSE	MAE	SSIM
0.1	0.0037	0.0180	0.9588
0.2	0.0047	0.0215	0.9443
0.3	0.0072	0.0294	0.8747

Required inference:
Explain how increasing corruption affects reconstruction quality and identify whether the denoising model is able to recover the major structure of the original image.







Detailed results (test set, 2,000 images; all metrics against the clean image).

Denoiser trained on	Test noise	Noisy MSE	Noisy SSIM	Den. MSE	Den. MAE	Den. SSIM
Gaussian $\sigma = 0.2$	Gaussian 0.1	0.0054	0.679	0.0037	0.0180	0.959
	Gaussian 0.2	0.0212	0.594	0.0047	0.0215	0.944
	Gaussian 0.3	0.0466	0.509	0.0072	0.0295	0.875
	S&P 0.05	0.0241	0.709	0.0049	0.0214	0.933
	S&P 0.10	0.0483	0.579	0.0071	0.0273	0.884
	S&P 0.20	0.0958	0.448	0.0134	0.0422	0.761
S&P $p = 0.10$	S&P 0.05	0.0241	0.709	0.0041	0.0191	0.956
	S&P 0.10	0.0483	0.579	0.0050	0.0214	0.944
	S&P 0.20	0.0958	0.448	0.0080	0.0288	0.889
	Gaussian 0.1	0.0054	0.679	0.0072	0.0298	0.877
	Gaussian 0.2	0.0212	0.594	0.0128	0.0485	0.698
	Gaussian 0.3	0.0466	0.509	0.0200	0.0701	0.603

Inference – Noise level vs. reconstruction quality and structure recovery.

Shows: As the noise level increases, the reconstruction quality decreases for both Gaussian and salt-and-pepper noise. For the Gaussian-trained denoising CAE, denoised MSE increases from 0.0037 to 0.0047 and 0.0072, while SSIM decreases from 0.959 to 0.944 and 0.875 as σ increases from 0.1 to 0.3. However, the denoised images consistently have much lower MSE and higher SSIM than the corresponding noisy inputs, showing that the model effectively removes a significant portion of the corruption.

A similar trend is observed for salt-and-pepper noise. The Gaussian-trained model achieves SSIM values of 0.933, 0.884, and 0.761 for noise levels 0.05, 0.10, and 0.20, respectively, while the S&P-trained model performs better on the same noise, achieving 0.956, 0.944, and 0.889. This shows that the denoising CAE performs best when tested on the same type of noise used during training.

At higher noise levels, some digit strokes and fine details are lost because the corruption makes the original structure harder to recover. Overall, the denoising CAE preserves the main structure of MNIST digits effectively under moderate corruption, with SSIM remaining above 0.94 for Gaussian noise up to $\sigma = 0.2$ when the model is trained on Gaussian noise. The results also show that the model is noise-specific, as performance decreases when the test noise differs from the training noise.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

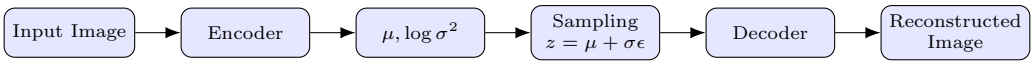
$$q_\phi(z|x) = \mathcal{N}\left(\mu(x), \mathrm{diag}(\sigma^2(x))\right).$$

The encoder produces:

$$\mu, \qquad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}\left(q_\phi(z|x) \parallel p(z)\right) = -\frac{1}{2}\sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.
Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

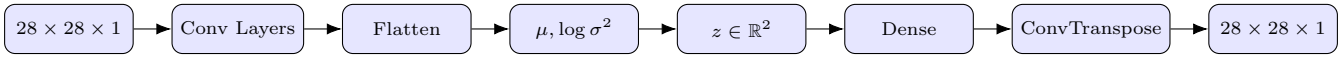
The encoder should produce:

$$\mu \in \mathbb{R}^2, \qquad \log \sigma^2 \in \mathbb{R}^2.$$

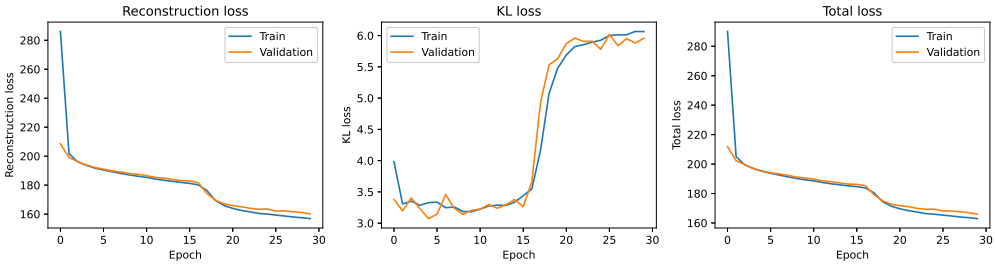
The decoder maps

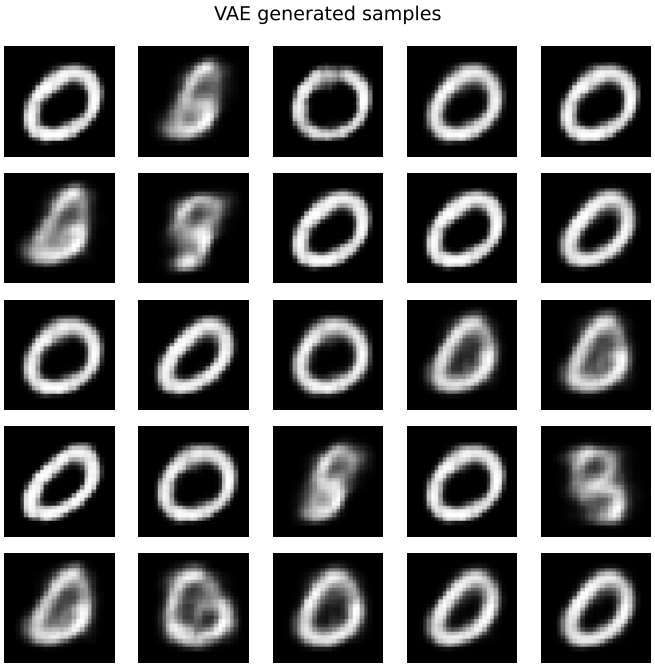
$$z \to 28 \times 28 \times 1.$$

A compact architecture is:



Implementation used. Encoder: Conv(32, stride 2) → Conv(64, stride 2) → Flatten → Dense(16) → $(\mu, \log \sigma^2)$ in $\mathbb{R}^2$ (69,076 parameters). Decoder: Dense($7 \cdot 7 \cdot 64$) → Reshape → ConvTranspose(64) → ConvTranspose(32) → ConvTranspose(1, sigmoid) (65,089 parameters). Total 134,165 parameters. Loss = per-image summed BCE + KL ($\beta = 1$), Adam 10^{-3}, batch 128, 30 epochs. Reconstruction metrics are computed from the decoded posterior mean μ.





Inference – VAE training and reconstruction. *Shows:*The VAE reconstruction loss decreases and stabilizes, while training and validation losses remain close, indicating no significant overfitting. On the test set, it achieves MSE = 0.0464 and SSIM = 0.440, with reconstructions appearing blurry and some digits losing their original shape. The limited 2-D latent bottleneck and KL regularization restrict the information stored in the latent space, leading to coarse reconstructions. Overall, the VAE trades reconstruction fidelity for a structured and sampleable latent representation.

18. Exploring the VAE Latent Space

Since the latent dimension is two, students can visualize the encoded test images in a two-dimensional plane.

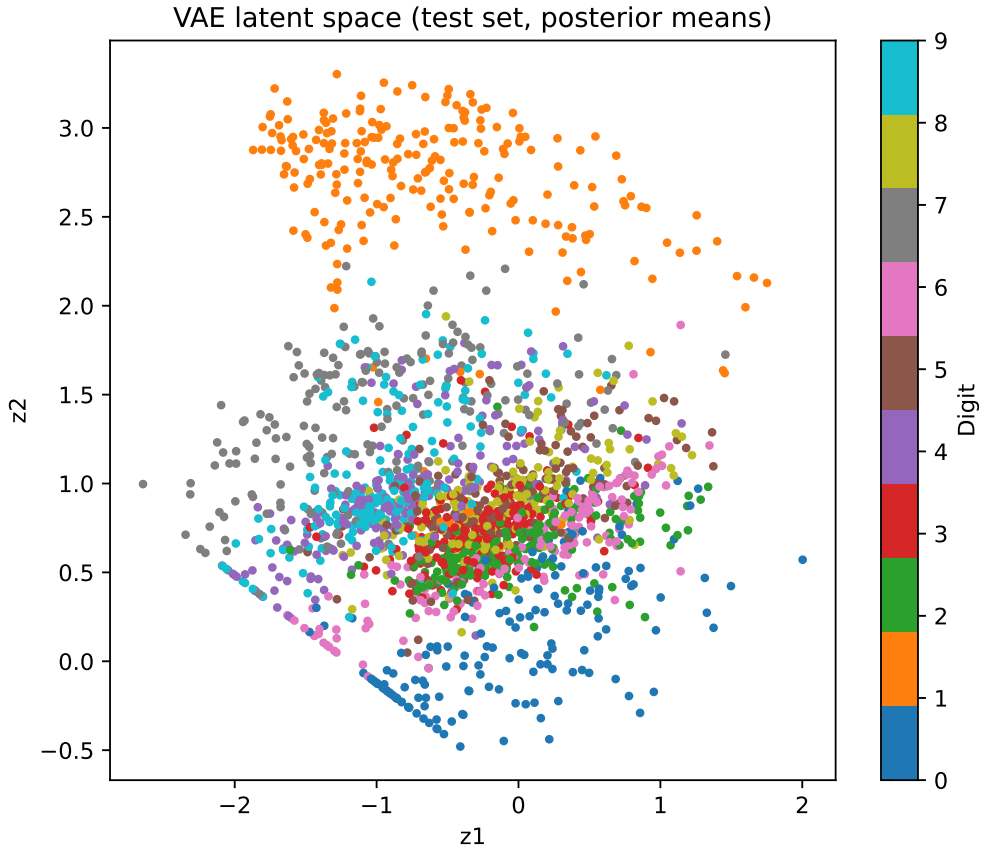
Plot 6: VAE Latent Space

- X-axis: z_1
- Y-axis: z_2
- Color/marker category: digit label

The labels are used only for visualization; they are not supplied to the VAE during training.

Required inference:

Students should describe whether samples belonging to the same digit occupy similar regions and whether different digits overlap.



Inference – VAE latent space. The 2-D latent representation forms compact clusters, with visually distinct digits such as 0 and 1 showing clearer separation, while similar digits such as 2, 3, 5 and 8 overlap. The KL regularization keeps the latent representations close to the prior, limiting the separation possible with only two dimensions. Overall, the latent space is organised mainly by visual similarity rather than class labels, as the labels were not used during training.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

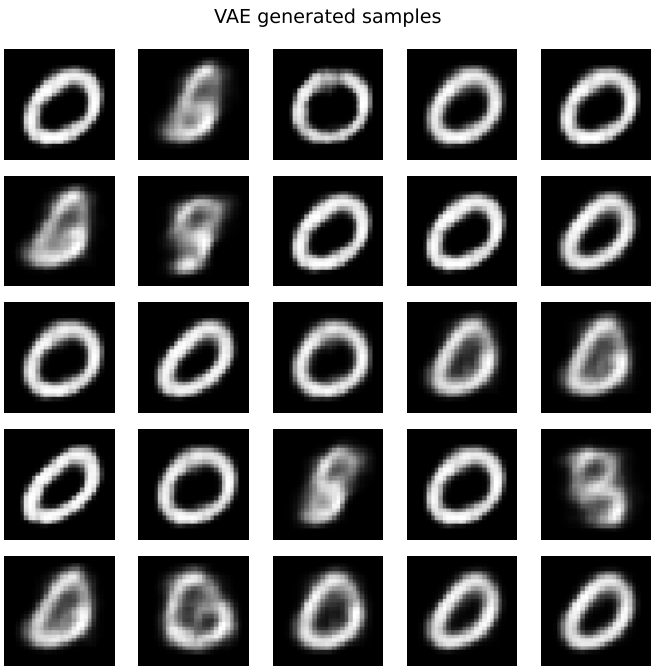
Plot 7: Randomly Generated Images

Generate at least 25 images using randomly sampled latent vectors.

Display them in a grid.

Students should determine:

- whether the generated images resemble handwritten digits;
- whether digit shapes are recognizable;
- whether there is visible variation among generated samples; and
- whether some samples are ambiguous or unrealistic.



20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \qquad z_B.$$

Construct

$$z(\alpha) = (1-\alpha)z_A + \alpha z_B, \qquad \alpha \in [0,1].$$

Use values such as:

$$\alpha = 0,\ 0.1,\ 0.2, \ldots, 1.$$

Decode every interpolated vector.

Plot 8: Latent Space Interpolation

Display the decoded images in sequence.

Required inference:

Describe whether the decoded image changes smoothly as the latent vector moves between two points.



21. Reconstruction Comparison Across Models

Compare the three reconstruction-based models:

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02049	0.05533	0.7581	211,040
Convolutional AE	0.00256	0.01500	0.9739	74,497
Denoising CAE ($\sigma = 0.2$)	0.00473	0.02151	0.9443	74,497

For the VAE, report its reconstruction loss separately because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss (summed BCE per image)	155.237
KL Loss	6.093
Total Loss	161.330
Test MSE	0.04603
Test MAE	0.10846
Mean SSIM	0.4466

22. Required Plots

The final report must contain the following plots:

1. Original image versus reconstructed image.
2. Training and validation loss for the basic autoencoder.
3. Fully connected AE versus convolutional AE reconstruction.
4. Clean versus noisy versus denoised images.
5. Noise level versus MSE/MAE/SSIM.
6. VAE latent-space visualization.
7. Randomly generated VAE images.
8. Latent-space interpolation.
9. Training/validation reconstruction loss for the VAE.
10. Reconstruction-error distribution on the test set.

For plots containing multiple quantitative measures, use separate figures or clearly labeled axes rather than placing incompatible scales on one axis.

23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

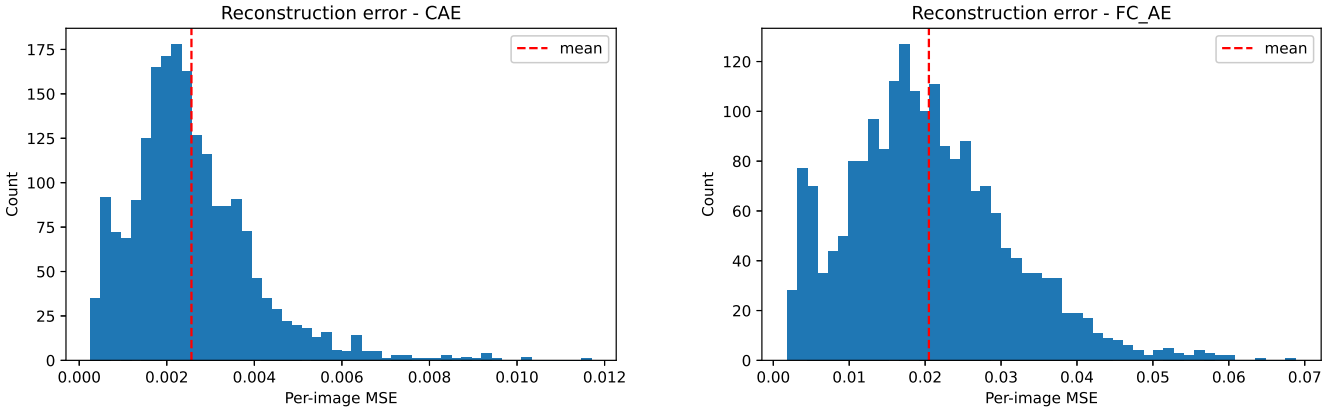
$$e_i = \frac{1}{D}\sum_{j=1}^{D}(x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: Distribution of Reconstruction Errors

Plot a histogram of e_i over the test set.

Students should identify:

- the typical reconstruction-error range;
- whether the distribution is concentrated or spread out;
- possible high-error samples; and
- representative images corresponding to high reconstruction error.



Plot 9: Distribution of per-image reconstruction error e_i on the 2,000 test images.

Model	Mean	Median	Std	5–95% range	Max
FC-AE	0.02049	0.01937	0.01063	0.0044–0.0394	0.06882
CAE	0.00256	0.00233	0.00141	0.00065–0.00516	0.01169

Inference – Error distribution. *Shows:* histograms of per-image MSE. *Trend:* both distributions are unimodal and slightly right-skewed, with most errors concentrated near the mean. Typical errors lie approximately in 0.0044–0.0394 for the FC-AE and 0.00065–0.00516 for the CAE, showing that the CAE errors are substantially lower. The FC-AE has a mean error of 0.02049, compared with only 0.00256 for the CAE, while the maximum errors are 0.06882 and 0.01169 respectively. *Why:* most digits have relatively simple structures and are reconstructed consistently, while a smaller number of difficult or unusual samples produce larger errors. *Conclusion:* the CAE provides consistently lower reconstruction error across the test set, and the error distribution supports the improved visual reconstruction quality observed in the CAE results.

24. Required High-Error Image Analysis

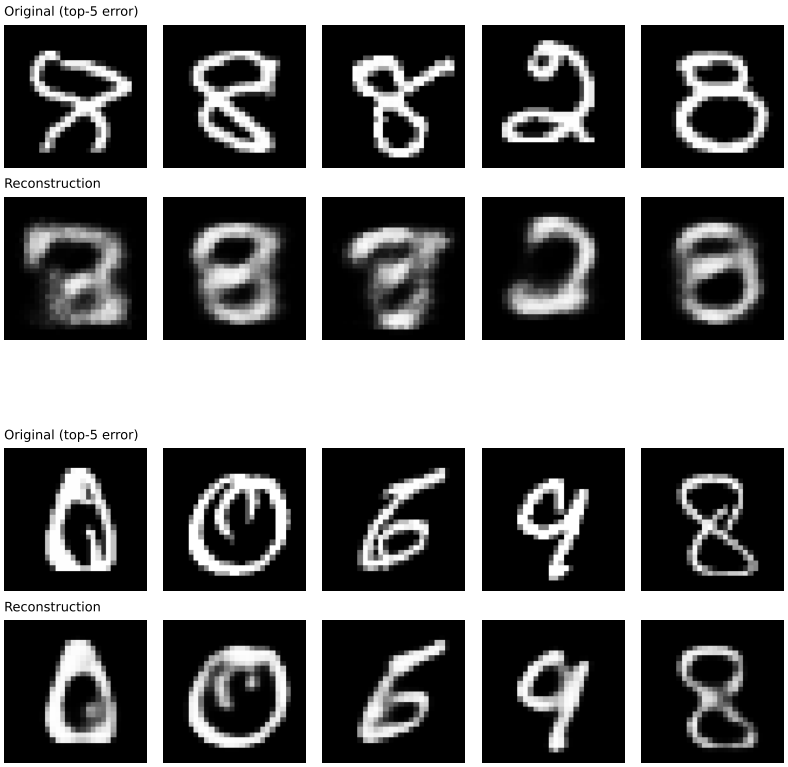
Select the five test images having the largest reconstruction errors.

Display:

	Fully Connected AE			Convolutional AE		
Sample	Test idx	Digit	Error e_i	Test idx	Digit	Error e_i
1	1782	8	0.0688	895	0	0.0117
2	1758	8	0.0637	1526	0	0.0102
3	1170	8	0.0607	1982	6	0.0102
4	744	2	0.0605	1594	9	0.0094
5	338	8	0.0589	431	8	0.0094

The original and reconstructed images for these samples are shown below (top row: original, bottom row: reconstruction; left to right = Sample 1 to 5).

Students should provide a short explanation for why these images may be difficult to reconstruct.



Inference – High-error images. Four of the five highest-error FC-AE samples are digit 8, while the remaining sample is digit 2. These digits contain unusual loops, thin strokes, or complex shapes that are difficult to represent accurately using the 16-D latent space, resulting in blurred or distorted reconstructions. For the CAE, the highest-error samples are digits 0, 6, 9, and 8; although their main structures remain recognisable, thin curves and additional internal strokes are partially lost. Overall, atypical writing styles, unusual strokes, and complex geometries produce higher reconstruction errors because the models learn compact representations of the more common digit structures.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time
FC Autoencoder	0.02049	0.05533	0.7581	211,040	16.00 s
Conv. Autoencoder	0.00256	0.01500	0.9739	74,497	37.08 s
Denoising CAE	0.00473	0.02151	0.9443	74,497	28.15 s
VAE ($d_z = 2$)	0.04603	0.10846	0.4466	134,165	40.89 s

The values must be obtained from the student's own execution.

Notes. FC-AE and CAE: reconstruction of the clean test image. Denoising CAE: reconstruction of the clean image from Gaussian $\sigma = 0.2$ noisy test input. VAE: metrics from the decoded posterior mean, 30 epochs, parameters = encoder + decoder. The times are wall-clock training times from this run (Colab).

26. Required Inferences

For every major plot, students must write a short inference containing:

1. What does the plot show?
2. What trend is observed?
3. Why might the observed trend occur?
4. What conclusion can be drawn?

The following inferences are mandatory:

- effect of latent dimension on reconstruction;
- difference between fully connected and convolutional reconstruction;
- effect of Gaussian noise level;
- ability of the denoising autoencoder to recover image structure;
- characteristics of the VAE latent distribution;
- structure of the two-dimensional VAE latent space;
- quality and diversity of generated samples;
- smoothness of latent-space interpolation;
- relationship between reconstruction error and visual quality; and
- differences between deterministic and probabilistic latent representations.

Mandatory inferences (summary of the experiment)

1. **Effect of latent dimension on reconstruction.** Test MSE decreases from 0.0460 ($d_z = 2$) to 0.0264 ($d_z = 8$), 0.0211 ($d_z = 16$), and 0.0202 ($d_z = 32$), while SSIM increases from 0.458 to 0.693, 0.748, and 0.761. The improvement becomes progressively smaller as the latent dimension increases, showing that a larger code preserves more information but provides diminishing reconstruction benefits.
2. **Fully connected vs. convolutional reconstruction.** The CAE achieves much lower MSE (0.0026 vs. 0.0205) and higher SSIM (0.974 vs. 0.758) than the FC-AE, while using only 74,497 parameters compared with 211,040. This improvement results from convolutional filters preserving local spatial relationships and image structure more effectively than fully connected layers.
3. **Effect of Gaussian noise level.** As σ increases from 0.1 to 0.3, the denoised MSE increases from 0.0037 to 0.0072, MAE increases from 0.0180 to 0.0295, and SSIM decreases from 0.959 to 0.875. Thus, increasing corruption gradually reduces reconstruction quality, although the denoiser continues to recover the main digit structure.
4. **Ability of the DAE to recover image structure.** For Gaussian noise with $\sigma = 0.2$, the noisy input has MSE 0.0212 and SSIM 0.594, while the denoised output achieves MSE 0.0047 and SSIM 0.944. This shows that the DAE effectively removes noise while preserving the main digit structure, although some fine strokes are slightly smoothed.
5. **Characteristics of the VAE latent distribution.** The 2-D VAE latent representation is compact and organised around the regularised latent prior. The posterior means have average values of approximately $(-0.44, 1.00)$ with standard deviations of about $(0.73, 0.72)$, while the average posterior standard deviations are small. The KL regularisation therefore encourages a compact and structured latent representation.
6. **Structure of the 2-D VAE latent space.** The latent representation shows clearer separation for visually distinct digits such as 0 and 1, while visually similar digits occupy overlapping regions. This indicates that the unsupervised latent space is organised primarily according to visual similarity rather than explicit class labels. The 2-D bottleneck is not large enough to clearly separate all ten digit classes.
7. **Quality and diversity of generated samples.** The generated samples are generally digit-like, but the reconstructions are relatively blurry and the generated outputs show limited diversity. This reflects the difficulty of representing the full variety of MNIST digits using only two latent dimensions and the regularisation imposed by the VAE.
8. **Smoothness of latent interpolation.** The latent interpolation produces gradual changes between the selected digit representations rather than abrupt transitions. This demonstrates that the VAE provides a continuous latent space in which neighbouring latent points can decode to visually related digit structures.
9. **Relationship between reconstruction error and visual quality.** Lower reconstruction error corresponds to better visual reconstruction quality. The CAE achieves MSE 0.00256 and SSIM 0.9739, compared with 0.02049 and 0.7581 for the FC-AE and 0.04603 and 0.4466 for the VAE. Within each model, the highest-error samples correspond to more difficult or unusual digit shapes, confirming the relationship between reconstruction error and visual distortion.
10. **Deterministic vs. probabilistic latent representations.** The autoencoders map each input to a deterministic latent representation and therefore focus directly on reconstruction accuracy. In contrast, the VAE represents each input using a probability distribution and regularises the latent space towards a structured prior, enabling sampling and interpolation. This additional structure comes at the cost of reconstruction fidelity, as reflected by the VAE's higher MSE and lower SSIM.

27. Additional Latent-Dimension Study

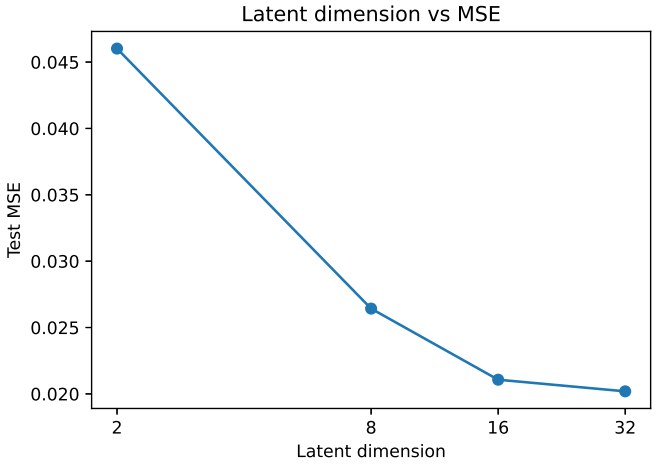
Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Report:

Latent Dimension	MSE	SSIM	Parameters
2	0.046016	0.458444	210,130
8	0.026425	0.692611	210,520
16	0.021068	0.747794	211,040
32	0.020192	0.761247	212,080

Plot 10: Latent Dimension vs. Reconstruction MSE
Students should explain the trade-off between compression and reconstruction quality.



Inference – Latent-dimension trade-off. *Shows:* test MSE and SSIM of the FC-AE for $d_z \in \{2, 8, 16, 32\}$. *Trend:* MSE decreases monotonically from 0.0460 to 0.0264, 0.0211, and 0.0202, while SSIM increases from 0.458 to 0.693, 0.748, and 0.761. The largest improvement occurs from $d_z = 2$ to $d_z = 8$, with progressively smaller gains as the latent dimension increases. The parameter count changes only slightly, from 210,130 to 212,080, because most parameters are in the 784×128 and 128×784 layers. *Why:* a small bottleneck cannot retain sufficient information about digit shape, stroke thickness, and variations in writing style, leading to greater reconstruction error. Increasing the latent dimension allows more information to be preserved, although the improvement becomes limited at higher dimensions. *Conclusion:* larger latent dimensions improve reconstruction quality, but with diminishing returns; $d_z = 16$ already captures most of the achievable reconstruction quality in this experiment.

28. Discussion Questions

1. **What is an autoencoder?**
   Answer: A neural network trained to reproduce its input at the output through a narrow bottleneck, so it learns a compact representation: $x \to z \to \hat{x}$.
2. **What is the purpose of the encoder?**
   Answer: To map the input image to a lower-dimensional latent representation $z = f_\theta(x)$ that retains the most important information.
3. **What is the purpose of the decoder?**
   Answer: To reconstruct the image from the latent code, $\hat{x} = g_\phi(z)$. In a VAE, it can also generate images from sampled latent codes.
4. **What is a latent representation?**
   Answer: The compressed hidden representation produced by the encoder; in this lab, the FC-AE uses 16 latent values in the main model, while the VAE uses 2 latent values.
5. **Why is the latent representation usually smaller than the input?**
   Answer: The bottleneck forces the network to learn useful structure instead of simply copying the input, enabling compression, feature learning, and noise removal.
6. **What happens when the latent dimension is too small?**
   Answer: Information is lost and reconstructions become less accurate. In the latent-dimension study, $d_z = 2$ gives MSE = 0.0460 and SSIM = 0.458, compared with MSE = 0.0202 and SSIM = 0.761 at $d_z = 32$.
7. **What happens when the latent dimension is increased?**
   Answer: Reconstruction quality improves, with MSE decreasing and SSIM increasing, but the improvement becomes smaller at higher dimensions. The parameter count also increases only slightly in this experiment.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?**
   Answer: Local receptive fields and shared weights exploit neighbouring-pixel relationships and spatial structure. In this experiment, the CAE achieved SSIM = 0.974 and MSE = 0.00256, compared with SSIM = 0.758 and MSE = 0.02049 for the FC-AE, while using fewer parameters.

9. **Differentiate an autoencoder and a denoising autoencoder.**
   Answer: A standard AE reconstructs its input, whereas a DAE receives a corrupted input $\tilde{x}$ and is trained to reconstruct the corresponding clean image x.
10. **Why is the clean image used as the target in denoising?**
    Answer: The clean image provides the desired output, so the network learns the mapping from noisy input to underlying image structure rather than reproducing the noise.
11. **What happens when the noise level is increased?**
    Answer: Both the noisy input and denoised output degrade as noise increases. For Gaussian noise, denoised MSE increases from 0.0037 to 0.0072 and SSIM decreases from 0.959 to 0.875 as σ increases from 0.1 to 0.3.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?**
    Answer: MSE measures squared pixel errors, MAE measures average absolute pixel error, and SSIM evaluates structural similarity. Together, they provide complementary measures of reconstruction quality.
13. **What is a Variational Autoencoder?**
    Answer: A generative autoencoder whose encoder outputs the parameters $(\mu, \log \sigma^2)$ of a latent distribution $q_\phi(z|x)$. A latent vector is sampled and decoded, while training minimises reconstruction loss together with KL divergence to a prior.
14. **How does a VAE differ from a conventional autoencoder?**
    Answer: A conventional AE uses a deterministic latent code and focuses on reconstruction, whereas a VAE uses a stochastic latent distribution, KL regularisation, and a structured latent space that can be sampled for generation.
15. **Why does a VAE learn a distribution instead of a single latent vector?**
    Answer: Learning a distribution makes the latent space structured and continuous, allowing nearby latent points to produce related outputs and enabling meaningful sampling and interpolation.
16. **What is the reparameterization trick?**
    Answer: It represents the latent sample as $z = \mu + \sigma \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. This separates the random sampling from the learnable parameters so gradients can propagate through μ and σ.
17. **Why is ϵ sampled from a standard normal distribution?**
    Answer: A standard normal provides a fixed noise source with zero mean and unit variance. Scaling and shifting it using μ and σ produces samples from the desired latent distribution.
18. **What is the role of KL divergence in the VAE loss?**
    Answer: KL divergence regularises the posterior towards the prior $p(z) = \mathcal{N}(0, I)$, helping create a structured and sampleable latent space. In the final test evaluation, the KL loss was 6.093 compared with a reconstruction loss of 155.237.
19. **What is the prior distribution used in the VAE?**
    Answer: The standard normal distribution, $p(z) = \mathcal{N}(0, I)$.
20. **Why is a two-dimensional latent space useful for visualization?**
    Answer: Two latent dimensions can be directly plotted as a scatter plot and decoded over a grid to visualize the organisation of the latent space and its corresponding image manifold.
21. **What does latent-space interpolation demonstrate?**
    Answer: It demonstrates the continuity of the latent space: moving between two latent codes produces gradual changes in the decoded images rather than completely unrelated outputs.
22. **Why can a VAE generate new images?**
    Answer: Because the latent space is regularised towards a known prior. New latent vectors can therefore be sampled from $\mathcal{N}(0, I)$ and passed through the decoder to generate new images.
23. **Why might some VAE-generated images be unclear?**
    Answer: The 2-D latent space provides limited capacity, while the reconstruction and KL objectives constrain the representation. As a result, generated samples can become blurry or ambiguous, particularly between visually similar digit regions.
24. **What is the difference between reconstruction and generation?**
    Answer: Reconstruction encodes a real image and decodes it back as $x \to z \to \hat{x}$, allowing comparison with the original. Generation samples a latent vector from the prior and decodes it to create an image without a corresponding ground-truth input.
25. **Why should test images remain unseen during training?**
    Answer: To provide an unbiased measure of generalisation. If test images were used during training, the evaluation could reflect memorisation rather than performance on unseen data. In this experiment, 2,000 test images were kept separate from the 9,000 training and 1,000 validation images.

29. Additional Exercises

Exercise 1: CAE with Different Latent Dimensions

Inference: Increasing the latent dimension allows the convolutional autoencoder to preserve more spatial and structural information from the input image. A larger latent representation can capture finer details such as digit shapes, stroke thickness, and variations in writing style. However, the improvement in reconstruction quality is expected to diminish at higher latent dimensions because most of the important information is already captured.

Exercise 2: Gaussian Noise vs. Salt-and-Pepper Noise

Inference: The denoising autoencoder performs best when the type of noise used during testing matches the noise used during training. The Gaussian-trained model achieves strong performance on Gaussian noise, while the salt-and-pepper-

trained model performs better on salt-and-pepper corruption. This demonstrates that the learned denoising behaviour is dependent on the statistical characteristics of the noise.

Exercise 3: Denoising with Different Noise Levels

Inference: As the noise level increases, the denoising task becomes progressively more difficult. For Gaussian noise, denoised MSE increases from 0.0037 at $\sigma = 0.10$ to 0.0047 at $\sigma = 0.20$ and 0.0072 at $\sigma = 0.30$, while SSIM decreases from 0.9588 to 0.9443 and 0.8748, respectively. This shows that stronger corruption results in greater reconstruction error and reduced structural similarity because more information about the original digit is lost.

Exercise 4: Transposed-Convolution Decoder

Inference: A transposed-convolution decoder allows the network to learn the upsampling operation rather than relying on fixed upsampling followed by convolution. This can improve the decoder's ability to reconstruct spatial details and produce sharper outputs. However, inappropriate kernel sizes or strides may introduce checkerboard artifacts, so the decoder architecture must be chosen carefully.

Exercise 5: VAE with Latent Dimensions 2 and 8

Inference: Increasing the VAE latent dimension provides greater capacity to represent variations in the input digits. A two-dimensional latent space is highly useful for visualization and reveals the organization of different digit classes, but its limited capacity can result in poorer reconstructions. An eight-dimensional latent space can retain more information and is therefore expected to improve reconstruction quality, although it is less directly interpretable visually.

Exercise 6: Increasing the KL-Loss Contribution

The VAE objective is given by

$$\mathcal{L}_{VAE} = \mathcal{L}_{reconstruction} + \beta \mathcal{L}_{KL}.$$

Inference: Increasing β places greater emphasis on regularizing the latent distribution towards a standard normal distribution. This produces a more structured and continuous latent space, which can improve the quality of sampling and interpolation. However, excessive KL regularization can reduce reconstruction fidelity because the model is encouraged to discard information from the latent representation.

Exercise 7: Larger VAE Sample Grid and Latent Interpolation

Inference: Generating a larger sample grid provides a broader view of the images learned by the VAE and helps evaluate whether the latent distribution produces meaningful digit-like samples. Latent interpolation between two digit regions demonstrates whether the learned representation changes smoothly. Smooth transitions indicate that the VAE has learned a continuous latent representation in which gradual changes in latent variables correspond to gradual changes in digit appearance.

30. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

$$\boxed{\text{Image} \rightarrow \text{Encoder} \rightarrow \text{Latent Representation} \rightarrow \text{Decoder} \rightarrow \text{Reconstructed Image}}$$

They should also demonstrate denoising:

$$\boxed{\text{Clean Image} \rightarrow \text{Noise} \rightarrow \text{Denoising Autoencoder} \rightarrow \text{Recovered Image}}$$

and variational generation:

$$\boxed{\text{Image} \rightarrow (\mu, \sigma) \rightarrow z \rightarrow \text{Decoder} \rightarrow \text{Generated/Reconstructed Image}}$$

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

31. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.

3. Pascal Vincent et al., “Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion,” Journal of Machine Learning Research, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, “MNIST Handwritten Digit Database.”
5. TensorFlow Documentation: `https://www.tensorflow.org`
6. Keras Documentation: `https://keras.io`
7. scikit-image Documentation: `https://scikit-image.org`